

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

January 2011

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(8 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Compute the stochastic differential of

$$Y(t) = \frac{(S_1(t))^2}{S_2(t)},$$

under the *historical probability measure* $\mathbf{P}$ and under the *risk-neutral measure* $\mathbf{Q}$. Your final result must be written in terms of the parameters ρ, v_1, v_2 .

Is Y a lognormal process? In case your answer is positive, compute the drift of Y under $\mathbf{P}$ and $\mathbf{Q}$, as well as its volatility.

Solution:

$$f(t, S_1, S_2) = \frac{(S_1)^2}{S_2}$$

The derivatives are

$$\begin{aligned}\frac{\partial f}{\partial t} &= 0, \quad \frac{\partial f}{\partial S_1} = \frac{2S_1}{S_2}, \quad \frac{\partial^2 f}{\partial S_1^2} = \frac{2}{S_2} \\ \frac{\partial f}{\partial S_2} &= -\frac{(S_1)^2}{(S_2)^2}, \quad \frac{\partial^2 f}{\partial S_2^2} = \frac{2(S_1)^2}{S_2^3}, \quad \frac{\partial^2 f}{\partial S_1 \partial S_2} = -\frac{2S_1}{S_2^2}\end{aligned}$$

and therefore

$$\begin{aligned}dY &= 0dt + \frac{2S_1}{S_2}dS_1 + \left(-\frac{(S_1)^2}{(S_2)^2}\right)dS_2 + \frac{1}{2}\left(\frac{2}{S_2}\right)\|\sigma_1\|^2 S_1^2 dt + \dots \\ &\quad + \frac{1}{2}\frac{2(S_1)^2}{S_2^3}\|\sigma_2\|^2 S_2^2 dt - \frac{2S_1}{S_2^2}\sigma_1 \cdot \sigma_2 S_1 S_2 dt \\ &= \frac{(S_1)^2}{S_2}((2\mu_1 - \mu_2 + \|\sigma_1\|^2 + \|\sigma_2\|^2 - 2\sigma_1 \cdot \sigma_2) dt + (2\sigma_1 - \sigma_2) \cdot dW^P(t)) \\ &= Y \left((2\mu_1 - \mu_2 + v_1^2 + v_2^2 - 2\rho v_1 v_2) dt + \left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2\right] dW^P(t) \right) \\ &= Y \left((r + v_1^2 + v_2^2 - 2\rho v_1 v_2) dt + \left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2\right] dW^Q(t) \right)\end{aligned}$$

where $\left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2\right] dW^P(t) \sim \left\| \left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2\right] \right\| dB(t)$
where B is a standard one dimensional P -Brownian motion.

In particular, $\left\| \left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2\right] \right\|^2 = (2v_1 - \rho v_2)^2 + \left(-\sqrt{1 - \rho^2} v_2\right)^2 = 4v_1^2 - 4\rho v_1 v_2 + v_2^2$

Y is lognormal both under P and Q with drift

$$\mu_Y^P = 2\mu_1 - \mu_2 + v_1^2 + v_2^2 - 2\rho v_1 v_2$$

and

$$\mu_Y^Q = r + v_1^2 + v_2^2 - 2\rho v_1 v_2$$

and volatility

$$\sigma_Y = \sqrt{4v_1^2 - 4\rho v_1 v_2 + v_2^2}$$

2. **(8 points)** Derive the *pathwise derivative estimate* for the Delta and the Vega of a call option in the Black-Scholes model. Does this method provide always an *unbiased* estimate of the greeks of a derivative? Can it always be applied?

3. **(7 points)** Derive the price of a perpetual American put option in the Black-Scholes model.
4. **(8 points)** A generic random variable X has cumulative distribution function F_X . Explain how to sample X with distribution function F_X by means of the inverse function method.

Consider now the random variable X whose cumulative distribution function is

$$F_X(x) = \begin{cases} \frac{x}{x+1} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Apply the inverse transform method to sample X from a uniform random variable.

Solution: The function F_X is strictly increasing for $x \geq 0$, since

$$F'_X(x) = \frac{x+1-x}{(x+1)^2} = \frac{1}{(x+1)^2} > 0$$

therefore can be inverted. Let $u \in (0; 1)$ then

$$\begin{aligned} F_X(x) &= u \\ \frac{x}{x+1} &= u \\ x &= u(x+1) \\ x(1-u) &= u \\ x &= \frac{u}{1-u} \\ F_X^{-1}(u) &= \frac{u}{1-u} \end{aligned}$$

Therefore if $U \sim \text{Uniform}(0; 1)$ by the inverse transform method $F_X^{-1}(U) = \frac{U}{1-U} \sim F_X$

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Good luck!

Questions:

1. **(7 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$.

Compute the stochastic differential of

$$Y(t) = \ln S_1(t) + \ln S_2(t)$$

with respect to the *risk neutral* probability measure. The diffusive coefficient of your final result must be written in terms of the parameters ρ, v_1, v_2 .

Solution:

$$f(t, S_1, S_2) = \ln S_1 + \ln S_2$$

The derivatives are

$$\begin{aligned} \frac{\partial f}{\partial t} &= 0, \quad \frac{\partial f}{\partial S_1} = \frac{1}{S_1}, \quad \frac{\partial^2 f}{\partial S_1^2} = -\frac{1}{S_1^2} \\ \frac{\partial f}{\partial S_2} &= \frac{1}{S_2}, \quad \frac{\partial^2 f}{\partial S_2^2} = -\frac{1}{S_2^2}, \quad \frac{\partial^2 f}{\partial S_1 \partial S_2} = 0 \end{aligned}$$

and therefore

$$\begin{aligned}
dY &= 0dt + \frac{1}{S_1}dS_1 + \frac{1}{S_2}dS_2 + \frac{1}{2} \left(-\frac{1}{S_1^2} \right) \|\sigma_1\|^2 S_1^2 dt + \frac{1}{2} \left(-\frac{1}{S_2^2} \right) \|\sigma_2\|^2 S_2^2 dt + 0 \\
&= \left(\mu_1 + \mu_2 - \frac{1}{2} \|\sigma_1\|^2 - \frac{1}{2} \|\sigma_2\|^2 \right) dt + (\sigma_1 + \sigma_2) \cdot dW^P(t) \\
&= \left(\mu_1 + \mu_2 - \frac{1}{2} \|\sigma_1\|^2 - \frac{1}{2} \|\sigma_2\|^2 \right) dt + \left[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^P(t) \\
&= \left(2r - \frac{1}{2} \|\sigma_1\|^2 - \frac{1}{2} \|\sigma_2\|^2 \right) dt + \left[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \\
&= \left(2r - \frac{1}{2} v_1^2 - \frac{1}{2} v_2^2 \right) dt + \left[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^Q(t)
\end{aligned}$$

where $\left[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \sim \left\| \left[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] \right\| dB(t)$
where B is a standard one dimensional Q -Brownian motion.

In particular, $\sigma_Y^2 = \left\| \left[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] \right\|^2 = (v_1 + \rho v_2)^2 + \left(\sqrt{1 - \rho^2} v_2 \right)^2 = v_1^2 + 2\rho v_1 v_2 + v_2^2$

2. **(4 points)** With respect to the notations of the previous point, compute the

$$\mathbb{Q} \left[Y(T) - Y\left(\frac{T}{2}\right) > 0 \right]$$

Your final result must be written in terms of the parameters r, ρ, v_1, v_2 and the cumulative distribution function of a standard normal random variable.

Solution: We know that

$$Y(t) = \ln S_1(0) + \ln S_2(0) + \left(2r - \frac{1}{2} \|\sigma_1\|^2 - \frac{1}{2} \|\sigma_2\|^2 \right) t + \sigma_Y B(t)$$

where B is a standard one dimensional Q -Brownian motion. Hence

$$\begin{aligned}
\mathbb{Q} \left[Y(T) - Y\left(\frac{T}{2}\right) > 0 \right] &= \mathbb{Q} \left[\left(2r - \frac{1}{2}v_1^2 - \frac{1}{2}v_2^2 \right) \frac{T}{2} + \sigma_Y \left(B(T) - B\left(\frac{T}{2}\right) \right) > 0 \right] \\
&= \mathbb{Q} \left[\sigma_Y \left(B(T) - B\left(\frac{T}{2}\right) \right) > - \left(2r - \frac{1}{2}v_1^2 - \frac{1}{2}v_2^2 \right) \frac{T}{2} \right] \\
&= \mathbb{Q} \left[\sigma_Y \sqrt{\frac{T}{2}} Z > - \left(2r - \frac{1}{2}v_1^2 - \frac{1}{2}v_2^2 \right) \frac{T}{2} \right] \\
&= \mathbb{Q} \left[Z > \frac{-(2r - \frac{1}{2}v_1^2 - \frac{1}{2}v_2^2)}{\sqrt{v_1^2 + 2\rho v_1 v_2 + v_2^2}} \sqrt{\frac{T}{2}} \right] = \mathcal{N} \left(\frac{2r - \frac{1}{2}v_1^2 - \frac{1}{2}v_2^2}{\sqrt{v_1^2 + 2\rho v_1 v_2 + v_2^2}} \sqrt{\frac{T}{2}} \right)
\end{aligned}$$

where $\mathcal{N}(\cdot)$ denotes the cumulative distribution function of a standard normal random variable.

3. **(8 points)** Explain (with details) how the control variate technique works in reducing the variance of a Monte Carlo estimate.
4. **(12 points)** In Merton's jump-diffusion model, write the stochastic differential equation satisfied by the risky security S (specify the assumptions on the coefficients and processes). State Ito formula for the jump-diffusive case (8 points), and apply it to derive the stochastic differential equation (4 points) of

$$(S(t))^2$$

Solution (computations only): We apply Ito formula

$$\begin{aligned}
d\varphi(t, S(t)) &= \frac{\partial \varphi(t; S(t^-))}{\partial t} dt + \frac{\partial \varphi(t; S(t^-))}{\partial S} \cdot S(t^-) (\mu dt + \sigma dW(t)) + \\
&\quad + \frac{1}{2} \frac{\partial^2 \varphi(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (\varphi(t, S(t)) - \varphi(t, S(t^-))) \Delta N(t)
\end{aligned}$$

to the function

$$\varphi(t, S) = S^2$$

Its derivatives are

$$\frac{\partial \varphi(t; S)}{\partial t} = 0 \quad \frac{\partial \varphi(t; S)}{\partial S} = 2S \quad \text{and} \quad \frac{\partial^2 \varphi(t; S)}{\partial S^2} = 2$$

Therefore we get

$$\begin{aligned}
d\varphi(t, S(t)) &= 0 \cdot dt + 2S(t^-) \cdot S(t^-) (\mu dt + \sigma dW(t)) + \\
&\quad + \frac{1}{2} 2 (S(t^-))^2 \sigma^2 dt + ((S(t))^2 - S(t^-)^2) \Delta N(t) \\
&= 2S(t^-) \cdot S(t^-) (\mu dt + \sigma dW(t)) + \frac{1}{2} 2 (S(t^-))^2 \sigma^2 dt + S(t^-)^2 (Y_j^2 - 1) \Delta N(t) \\
&= (S(t^-))^2 \cdot [(2\mu + \sigma^2) dt + 2\sigma dW(t) + (Y_j^2 - 1) \Delta N(t)]
\end{aligned}$$

Hence

$$\frac{d(S(t))^2}{(S(t^-))^2} = (2\mu + \sigma^2) dt + 2\sigma dW(t) + (Y_j^2 - 1) \Delta N(t)$$

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Questions:

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for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$.

Compute the stochastic differential of

$$Y(t) = S_1(t) \cdot (S_2(t))^3$$

with respect to the *risk neutral* probability measure. The diffusive coefficient of your final result must be written in terms of the parameters ρ, v_1, v_2 .

Solution:

$$f(t, S_1, S_2) = S_1 \cdot S_2^3$$

The derivatives are

$$\begin{aligned} \frac{\partial f}{\partial t} &= 0, \quad \frac{\partial f}{\partial S_1} = S_2^3, \quad \frac{\partial^2 f}{\partial S_1^2} = 0 \\ \frac{\partial f}{\partial S_2} &= 3S_1 \cdot S_2^2, \quad \frac{\partial^2 f}{\partial S_2^2} = 6S_1 \cdot S_2, \quad \frac{\partial^2 f}{\partial S_1 \partial S_2} = 3S_2^2 \end{aligned}$$

and therefore

$$\begin{aligned}
dY &= 0dt + S_2^3 dS_1 + 3S_1 \cdot S_2^2 dS_2 + \frac{1}{2} (6S_1 \cdot S_2) \|\sigma_2\|^2 S_2^2 dt + 3S_2^2 \sigma_1 \cdot \sigma_2 S_1 S_2 dt \\
&= S_1 \cdot S_2^3 \cdot \left[(\mu_1 + 3\mu_2 + 3\|\sigma_2\|^2 + 3\sigma_1 \cdot \sigma_2) dt + (\sigma_1 + 3\sigma_2) \cdot dW^P(t) \right] \\
&= S_1 \cdot S_2^3 \cdot \left[(4r + 3v_2^2 + 3\rho v_1 v_2) dt + \left[v_1 + 3\rho v_2, 3\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right]
\end{aligned}$$

where $\left[v_1 + 3\rho v_2, 3\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \sim \left\| \left[v_1 + 3\rho v_2, 3\sqrt{1 - \rho^2} v_2 \right] \right\| dB(t)$
where B is a standard one dimensional Q -Brownian motion.

In particular, $\sigma_Y^2 = \left\| \left[v_1 + 3\rho v_2, 3\sqrt{1 - \rho^2} v_2 \right] \right\|^2 = v_1^2 + 6\rho v_1 v_2 + 9v_2^2$

2. **(4 points)** With respect to the notations of the previous point, compute the

$$\mathbb{Q} \left[\frac{Y(T)}{Y\left(\frac{T}{2}\right)} > 1 \right]$$

Your final result must be written in terms of the parameters r, ρ, v_1, v_2 and the cumulative distribution function of a standard normal random variable.

Solution: We know that Y is lognormal, with $\mu_Y^{\mathbb{Q}} = 4r + 3v_2^2 + 3\rho v_1 v_2$ and volatility σ_Y . Hence

$$Y(t) = Y(0) \cdot \exp \left(\left(\mu_Y^{\mathbb{Q}} - \frac{1}{2} \sigma_Y^2 \right) t + \sigma_Y B(t) \right)$$

where B is a standard one dimensional Q -Brownian motion. Hence

$$\begin{aligned}
\mathbb{Q} \left[\frac{Y(T)}{Y\left(\frac{T}{2}\right)} > 1 \right] &= \mathbb{Q} \left[\exp \left(\left(\mu_Y^{\mathbb{Q}} - \frac{1}{2} \sigma_Y^2 \right) \frac{T}{2} + \sigma_Y \left(B(T) - B\left(\frac{T}{2}\right) \right) \right) > 1 \right] \\
&= \mathbb{Q} \left[\sigma_Y \left(B(T) - B\left(\frac{T}{2}\right) \right) > - \left(\mu_Y^{\mathbb{Q}} - \frac{1}{2} \sigma_Y^2 \right) \frac{T}{2} \right] \\
&= \mathbb{Q} \left[\sigma_Y \sqrt{\frac{T}{2}} Z > - \left(\mu_Y^{\mathbb{Q}} - \frac{1}{2} \sigma_Y^2 \right) \frac{T}{2} \right] \\
&= \mathbb{Q} \left[Z > - \frac{4r + 3v_2^2 + 3\rho v_1 v_2 - 0.5 \cdot (v_1^2 + 6\rho v_1 v_2 + 9v_2^2)}{\sqrt{v v_1^2 + 6\rho v_1 v_2 + 9v_2^2}} \sqrt{\frac{T}{2}} \right] = \mathcal{N} \left(\frac{-0.5v_1^2 - 1.5v_2^2 + 4r}{\sqrt{v v_1^2 + 6\rho v_1 v_2 + 9v_2^2}} \right)
\end{aligned}$$

where $\mathcal{N}(\cdot)$ denotes the cumulative distribution function of a standard normal random variable.

3. **(8 points)** Explain (with details) how and when the antithetic variable technique works in reducing the variance of a Monte Carlo estimate.
4. **(12 points)** In the discrete-time framework consider a path-dependent European lookback option on the underlying S , whose payoff at date t is

$$X(t) = \max_{0 \leq s \leq t} S(s) - S(t).$$

Explain how to Markovianize the pricing problem for the lookback option (5 points).

Explain the main features of the forward shooting grid method that can be employed to reduce the computational burden of the pricing algorithm (7 points).

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for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the probability of the event

$$(S_1(T))^2 > S_2(T)$$

using both the *historical probability measure* $\mathbf{P}$ and the *risk-neutral probability measure* $\mathbf{Q}$. (The argument of the cumulative normal distribution function of your final result must be written in terms of the parameters ρ, v_1, v_2).

Solution: The event

$$(S_1(T))^2 > S_2(T)$$

whose probability we are interested in can be rewritten as

$$\begin{aligned}
\left(S_1(0) \cdot e^{\left(\mu_1 - \frac{\|\sigma_1\|^2}{2} \right) T + \sigma_1 W^P(T)} \right)^2 &> S_2(0) \cdot e^{\left(\mu_2 - \frac{\|\sigma_2\|^2}{2} \right) T + \sigma_2 W^P(T)} \\
e^{2 \left(\mu_1 - \frac{\|\sigma_1\|^2}{2} \right) T + 2\sigma_1 W^P(T)} &> e^{\left(\mu_2 - \frac{\|\sigma_2\|^2}{2} \right) T + \sigma_2 W^P(T)} \\
2 \left(\mu_1 - \frac{\|\sigma_1\|^2}{2} \right) T + 2\sigma_1 W^P(T) &> \left(\mu_2 - \frac{\|\sigma_2\|^2}{2} \right) T + \sigma_2 W^P(T) \\
(2\sigma_1 - \sigma_2) W^P(T) &> \left(-2\mu_1 + \|\sigma_1\|^2 + \mu_2 - \frac{\|\sigma_2\|^2}{2} \right) T
\end{aligned}$$

with particular reference with respect to the *historical probability measure* $\mathbf{P}$. Hence the *historical probability* we have to compute is

$$\mathbf{P} \left((2\sigma_1 - \sigma_2) W^P(T) > \left(-2\mu_1 + \|\sigma_1\|^2 + \mu_2 - \frac{\|\sigma_2\|^2}{2} \right) T \right).$$

We have that

$$\left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2 \right] W^P(T) \stackrel{\mathbf{P}}{\sim} \sigma_Y B(T)$$

where B is a standard one dimensional $\mathbf{P}$ -Brownian motion, and

$$\begin{aligned}
\sigma_Y^2 &= \left\| \left[2v_1 - \rho v_2, -\sqrt{1 - \rho^2} v_2 \right] \right\|^2 = \\
&= (2v_1 - \rho v_2)^2 + \left(-\sqrt{1 - \rho^2} v_2 \right)^2 = 4v_1^2 - 4\rho v_1 v_2 + v_2^2.
\end{aligned}$$

Hence, denoting with

$$\begin{aligned}
\mu_Y^P &= -2\mu_1 + \|\sigma_1\|^2 + \mu_2 - \frac{\|\sigma_2\|^2}{2} \\
&= -2\mu_1 + v_1^2 + \mu_2 - \frac{v_2^2}{2}
\end{aligned}$$

we obtain

$$\begin{aligned}
& \mathbf{P} \left((2\sigma_1 - \sigma_2) W^P(T) > \left(-2\mu_1 + \|\sigma_1\|^2 + \mu_2 - \frac{\|\sigma_2\|^2}{2} \right) T \right) = \mathbf{P} (\sigma_Y \cdot B(T) > \mu_Y^P \cdot T) \\
& = \mathbf{P} (\sigma_Y \cdot \sqrt{T} Z > \mu_Y^P \cdot T) = \mathbf{P} \left(Z > \frac{\mu_Y^P}{\sigma_Y} \cdot \sqrt{T} \right) \quad \text{where } Z \stackrel{\mathbf{P}}{\sim} N(0, 1) \\
& = \mathbf{P} \left(Z < -\frac{\mu_Y^P}{\sigma_Y} \cdot \sqrt{T} \right) = \mathbf{P} \left(Z < \frac{+2\mu_1 - v_1^2 - \mu_2 + \frac{v_2^2}{2}}{\sqrt{4v_1^2 - 4\rho v_1 v_2 + v_2^2}} \cdot \sqrt{T} \right) = \\
& = N \left(\frac{+2\mu_1 - v_1^2 - \mu_2 + \frac{v_2^2}{2}}{\sqrt{4v_1^2 - 4\rho v_1 v_2 + v_2^2}} \cdot \sqrt{T} \right)
\end{aligned}$$

Similarly,

$$\mathbf{Q} (S_1(T)^2 > S_2(T)) = \mathbf{Q} \left(\left(S_1(0) \cdot e^{\left(r - \frac{\|\sigma_1\|^2}{2} \right) T + \sigma_1 W^Q(T)} \right)^2 > S_2(0) \cdot e^{\left(r - \frac{\|\sigma_2\|^2}{2} \right) T + \sigma_2 W^Q(T)} \right)$$

where W^Q is a 2-dimensional standard Brownian motion with respect to the *risk-neutral probability* $\mathbf{Q}$. Therefore

$$\begin{aligned}
\mathbf{Q} (S_1(T)^2 > S_2(T)) & = \mathbf{Q} \left(\left(e^{\left(r - \frac{\|\sigma_1\|^2}{2} \right) T + \sigma_1 W^Q(T)} \right)^2 > e^{\left(r - \frac{\|\sigma_2\|^2}{2} \right) T + \sigma_2 W^Q(T)} \right) \\
& = \mathbf{Q} \left((2\sigma_1 - \sigma_2) W^Q(T) > \left(-2r + \|\sigma_1\|^2 + r - \frac{\|\sigma_2\|^2}{2} \right) T \right) \\
& = \mathbf{Q} \left(\sigma_Y \cdot \sqrt{T} Z^Q > \left(-r + \|\sigma_1\|^2 - \frac{\|\sigma_2\|^2}{2} \right) T \right) \quad \text{where } Z^Q \stackrel{\mathbf{Q}}{\sim} N(0, 1) \\
& = \mathbf{Q} \left(Z^Q > \frac{\left(-r + v_1^2 - \frac{v_2^2}{2} \right)}{\sigma_Y} \sqrt{T} \right) \\
& = \mathbf{Q} \left(Z^Q < -\frac{\left(-r + v_1^2 - \frac{v_2^2}{2} \right)}{\sigma_Y} \sqrt{T} \right) = N \left(\frac{r - v_1^2 + \frac{v_2^2}{2}}{\sqrt{4v_1^2 - 4\rho v_1 v_2 + v_2^2}} \sqrt{T} \right).
\end{aligned}$$

2. **(8 points)** Derive the *pathwise derivative estimate* for the Delta and the Vega of a call option in the Black-Scholes model. Does this method

provide always an *unbiased* estimate of the greeks of a derivative? Can it always be applied?

3. **(8 points)** In Merton's jump-diffusion model, write the stochastic differential equation satisfied by the risky security S with respect to the *historical probability measure* $\mathbf{P}$ (specify the assumptions on the coefficients and processes). State Ito formula for a deterministic function of time t and of the jump-diffusion process.
4. **(7 points)** Use Ito formula for jump diffusions to compute the stochastic differential of

$$X(t) = e^{\alpha t} \cdot (S(t))^3$$

where S is the risky asset in Merton's jump diffusion model, and $\alpha \in \mathbb{R}$. Is the process X a jump-diffusion?

Solution: The derivatives of the function

$$\varphi(t, S) = e^{\alpha t} (S)^3$$

are

$$\frac{\partial \varphi}{\partial t} = \alpha e^{\alpha t} (S)^3, \quad \frac{\partial \varphi}{\partial S} = 3e^{\alpha t} S^2, \quad \frac{\partial^2 \varphi}{\partial S^2} = 6e^{\alpha t} S$$

Therefore

$$\begin{aligned} d(e^{\alpha t} \cdot (S(t))^3) &= d\varphi(t, S(t)) = \frac{\partial \varphi(t; S(t^-))}{\partial t} dt + \frac{\partial \varphi(t; S(t^-))}{\partial S} \cdot S(t^-) (\mu dt + \sigma dW(t)) + \\ &\quad + \frac{1}{2} \frac{\partial^2 \varphi(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (\varphi(t, S(t)) - \varphi(t, S(t^-))) \Delta N(t) \\ &= \alpha e^{\alpha t} (S(t^-))^3 dt + 3e^{\alpha t} S^2(t^-) \cdot S(t^-) (\mu dt + \sigma dW(t)) + \\ &\quad + \frac{1}{2} 6e^{\alpha t} S(t^-) (S(t^-))^2 \sigma^2 dt + \left(e^{\alpha t} (S(t))^3 - e^{\alpha t} (S(t^-))^3 \right) \Delta N(t) \\ &= e^{\alpha t} (S(t^-))^3 \cdot [\alpha dt + 3(\mu dt + \sigma dW(t)) + 3\sigma^2 dt] + \left(e^{\alpha t} (S(t^-) Y_j)^3 - e^{\alpha t} (S(t^-))^3 \right) \Delta N(t) \\ &= e^{\alpha t} (S(t^-))^3 \cdot [\alpha dt + 3(\mu dt + \sigma dW(t)) + 3\sigma^2 dt + (Y_j^3 - 1) \Delta N(t)] \\ &= e^{\alpha t} (S(t^-))^3 \cdot [(\alpha + 3\mu + 3\sigma^2) dt + 3\sigma dW(t) + (Y_j^3 - 1) \Delta N(t)] \end{aligned}$$

that is

$$\frac{d(e^{\alpha t} \cdot (S(t))^3)}{e^{\alpha t} (S(t^-))^3} = (\alpha + 3\mu + 3\sigma^2) dt + 3\sigma dW(t) + (Y_j^3 - 1) \Delta N(t)$$

or equivalently

$$\frac{d \left(e^{\alpha t} \cdot (S(t))^3 \right)}{e^{\alpha t} (S(t^-))^3} = (\alpha + 3\mu + 3\sigma^2) dt + 3\sigma dW(t) + dJ_\varphi(t)$$

where J_φ is the compound Poisson process defined as follows

$$J_\varphi(t) = \sum_{j=1}^{N(t)} (Y_j^3 - 1),$$

with *jump size* $Y_j^3 - 1$, i.i.d. Hence $X(t) = e^{\alpha t} \cdot (S(t))^3$ is a jump-diffusion process, with *drift* $\alpha + 3\mu + 3\sigma^2$, and *volatility* 3σ .

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

February 2012

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(8 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the stochastic differential of

$$Y(t) = (S_1(t))^{\frac{1}{3}} \cdot S_2(t)$$

with respect to the *risk neutral* and the *historical* probability measure.

Is the process Y lognormal? If your answer is positive, compute the drift and the volatility of Y with respect to the *risk neutral* and the *historical* probability measure. (*Your final results must be written in terms of the parameters ρ, v_1, v_2 .*)

Solution:

$$f(t, S_1, S_2) = (S_1)^{\frac{1}{3}} \cdot S_2$$

The derivatives are

$$\begin{aligned}\frac{\partial f}{\partial t} &= 0, \quad \frac{\partial f}{\partial S_1} = \frac{1}{3} \cdot (S_1)^{-\frac{2}{3}} \cdot S_2, \quad \frac{\partial^2 f}{\partial S_1^2} = -\frac{2}{9} \cdot (S_1)^{-\frac{5}{3}} \cdot S_2 \\ \frac{\partial f}{\partial S_2} &= (S_1)^{\frac{1}{3}}, \quad \frac{\partial^2 f}{\partial S_2^2} = 0, \quad \frac{\partial^2 f}{\partial S_1 \partial S_2} = \frac{1}{3} \cdot (S_1)^{-\frac{2}{3}}\end{aligned}$$

and therefore

$$\begin{aligned}dY &= 0dt + \frac{1}{3} \cdot (S_1)^{-\frac{2}{3}} \cdot S_2 dS_1 + (S_1)^{\frac{1}{3}} dS_2 + \frac{1}{2} \left(-\frac{2}{9} \cdot (S_1)^{-\frac{5}{3}} \cdot S_2 \right) \|\sigma_1\|^2 S_1^2 dt + \\ &\quad + 0 + \left(\frac{1}{3} \cdot (S_1)^{-\frac{2}{3}} \right) \sigma_1 \cdot \sigma_2 S_1 S_2 dt + 0 \\ &= (S_1)^{\frac{1}{3}} S_2 \left\{ \left(\frac{1}{3} \mu_1 + \mu_2 - \frac{1}{9} \|\sigma_1\|^2 + \frac{1}{3} \sigma_1 \cdot \sigma_2 \right) dt + \left(\frac{1}{3} \sigma_1 + \sigma_2 \right) \cdot dW^P(t) \right\} \\ &= Y \left\{ \left(\frac{1}{3} \mu_1 + \mu_2 - \frac{1}{9} v_1^2 + \frac{1}{3} \rho v_1 v_2 \right) dt + \left[\frac{1}{3} v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^P(t) \right\} \\ &= Y \left\{ \left(\frac{1}{3} r + r - \frac{1}{9} v_1^2 + \frac{1}{3} \rho v_1 v_2 \right) dt + \left[\frac{1}{3} v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\} \\ &= Y \left\{ \left(\frac{4}{3} r - \frac{1}{9} v_1^2 + \frac{1}{3} \rho v_1 v_2 \right) dt + \left[\frac{1}{3} v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\}\end{aligned}$$

where $\left[\frac{1}{3} v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \sim \left\| \left[\frac{1}{3} v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] \right\| dB(t)$
where B is a standard one dimensional Q -Brownian motion.

The process Y is lognormal with respect to both the *risk neutral* and the *historical* probability measure, with volatility σ_Y given by

$$\sigma_Y^2 = \left\| \left[\frac{1}{3} v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2 \right] \right\|^2 = \left(\frac{1}{3} v_1 + \rho v_2 \right)^2 + \left(\sqrt{1 - \rho^2} v_2 \right)^2 = \frac{1}{9} v_1^2 + \frac{2}{3} \rho v_1 v_2 + v_2^2$$

and drift

$$\begin{aligned}&\frac{1}{3} \mu_1 + \mu_2 - \frac{1}{9} v_1^2 + \frac{1}{3} \rho v_1 v_2 \quad \text{with respect to the historical probability } P \\ &\frac{4}{3} r - \frac{1}{9} v_1^2 + \frac{1}{3} \rho v_1 v_2 \quad \text{with respect to the risk neutral probability } Q.\end{aligned}$$

2. **(5 points)** In the discrete-time framework state the backward recursive formula to compute the price for an American *put* option on a risky asset S .

3. **(8 points)** Explain (with details) how the control variate technique works in reducing the variance of a Monte Carlo estimate. When is this technique effective to reduce the variance? Does it provide an *unbiased* estimate?
4. **(10 points)** In the discrete-time framework consider a path-dependent European lookback option on the underlying S , whose payoff at date t is

$$X(t) = \max_{0 \leq s \leq t} S(s) - S(t).$$

Explain how to Markovianize the pricing problem for the lookback option (4 points).

Explain the main features of the forward shooting grid method that can be employed to reduce the computational burden of the pricing algorithm (6 points).

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

September 2012

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(12 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Suppose

$$Y(T) = (\ln S_1(T) + \ln S_2(T))^2$$

is the terminal payoff of an European derivative. Compute its initial no-arbitrage price **(8 points)**.

Does the initial price of the derivative increase or decrease if ρ , the correlation between the two assets S_1 and S_2 , increases? **(4 points)**.

(Your final results must be written in terms of the parameters ρ, v_1, v_2).

Solution: The initial no-arbitrage price of the derivative is

$$\mathbb{E}^{\mathbb{Q}} [e^{-rT} Y(T)]$$

and therefore we compute

$$\begin{aligned}
& \mathbb{E}^{\mathbb{Q}} \left[(\ln S_1(T) + \ln S_2(T))^2 \right] = \\
&= \mathbb{E}^{\mathbb{Q}} \left[\left(\left(r - \frac{\|\sigma_1\|^2}{2} \right) T + \sigma_1 W^Q(T) + \left(r - \frac{\|\sigma_2\|^2}{2} \right) T + \sigma_2 W^Q(T) \right)^2 \right] = \\
&= \mathbb{E}^{\mathbb{Q}} \left[\left(\left(2r - \frac{\|\sigma_1\|^2 + \|\sigma_2\|^2}{2} \right) T + (\sigma_1 + \sigma_2) W^Q(T) \right)^2 \right] = \\
&= \mathbb{E}^{\mathbb{Q}} \left[\left(\left(2r - \frac{v_1^2 + v_2^2}{2} \right) T + [v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] W^Q(T) \right)^2 \right] = \\
&= \mathbb{E}^{\mathbb{Q}} \left[\left(2r - \frac{v_1^2 + v_2^2}{2} \right)^2 T^2 \right] + \mathbb{E}^{\mathbb{Q}} \left[\left([v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] W^Q(T) \right)^2 \right] + \\
&\quad + \mathbb{E}^{\mathbb{Q}} \left[2 \left(2r - \frac{v_1^2 + v_2^2}{2} \right) T [v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] W^Q(T) \right]
\end{aligned}$$

where $[v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] W^Q(T) \sim \left\| [v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] \right\| \sqrt{T} Z$ with Z standard normal random variable w.r.t. $\mathbb{Q}$. Since $\left\| [v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] \right\| = \sqrt{v_1^2 + 2\rho v_1 v_2 + v_2^2}$, we have that the second addend in the last sum is

$$\begin{aligned}
& \mathbb{E}^{\mathbb{Q}} \left[\left([v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] W^Q(T) \right)^2 \right] = \\
&= \mathbb{E}^{\mathbb{Q}} \left[\left(\sqrt{v_1^2 + 2\rho v_1 v_2 + v_2^2} \sqrt{T} Z \right)^2 \right] \\
&= T (v_1^2 + 2\rho v_1 v_2 + v_2^2) \cdot \mathbb{E}^{\mathbb{Q}} [Z^2] \\
&= T (v_1^2 + 2\rho v_1 v_2 + v_2^2) \cdot 1
\end{aligned}$$

The last term in the sum is null because

$$\begin{aligned}
& \mathbb{E}^{\mathbb{Q}} \left[2 \left(2r - \frac{v_1^2 + v_2^2}{2} \right) T [v_1 + \rho v_2, \sqrt{1 - \rho^2} v_2] W^Q(T) \right] = \\
&= 2 \left(2r - \frac{v_1^2 + v_2^2}{2} \right) T \sqrt{v_1^2 + 2\rho v_1 v_2 + v_2^2} \sqrt{T} \mathbb{E}^{\mathbb{Q}} [Z] = 0,
\end{aligned}$$

since $\mathbb{E}^{\mathbb{Q}}[Z] = 0$. Hence the initial no-arbitrage price of the derivative is

$$\mathbb{E}^{\mathbb{Q}}[e^{-rT}Y(T)] = e^{-rT} \left[\left(2r - \frac{v_1^2 + v_2^2}{2} \right)^2 T^2 + (v_1^2 + 2\rho v_1 v_2 + v_2^2) T \right]$$

The no-arbitrage price of the derivative is increasing w.r.t. ρ since its derivative is

$$\frac{\partial}{\partial \rho} \mathbb{E}^{\mathbb{Q}}[e^{-rT}Y(T)] = e^{-rT} 2v_1 v_2 T > 0.$$

2. **(6 points)** In the multidimensional Black-Scholes model define the market price of risk vector. Explain how market completeness and no-arbitrage are related to the market price of risk vector.
3. **(5 points)** Explain how the antithetic variate technique works in reducing the variance of a Monte Carlo estimate. When is this technique effective to reduce the variance? Does it provide an *unbiased* estimate?
4. **(8 points)** In the standard Black-Scholes framework derive the no-arbitrage price of an American perpetual *put* option on the lognormal asset S .

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

January 2013

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(9 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the probability of the event

$$S_1(T) > S_1\left(\frac{T}{2}\right) \cdot S_2\left(\frac{T}{2}\right)$$

using both the *historical probability measure* $\mathbb{P}$ and the *risk-neutral probability measure* $\mathbb{Q}$. (The argument of the cumulative normal distribution function of your final result must be written in terms of the parameters ρ, v_1, v_2).

Solution: The event

$$S_1(T) > S_1\left(\frac{T}{2}\right) \cdot S_2\left(\frac{T}{2}\right)$$

whose probability we are interested in can be rewritten as

$$\begin{aligned}
& S_1(0) \cdot e^{\left(\mu_1 - \frac{\|\sigma_1\|^2}{2}\right)T + \sigma_1 W^P(T)} > S_1(0) \cdot e^{\left(\mu_1 - \frac{\|\sigma_1\|^2}{2}\right)\left(\frac{T}{2}\right) + \sigma_1 W^P\left(\frac{T}{2}\right)} \cdot S_2(0) \cdot \\
& e^{\left(\mu_2 - \frac{\|\sigma_2\|^2}{2}\right)\left(\frac{T}{2}\right) + \sigma_2 W^P\left(\frac{T}{2}\right)} \\
& e^{\left(\mu_1 - \frac{\|\sigma_1\|^2}{2}\right)T + \sigma_1 W^P(T)} > e^{\left(\mu_1 - \frac{\|\sigma_1\|^2}{2}\right)\left(\frac{T}{2}\right) + \sigma_1 W^P\left(\frac{T}{2}\right) + \left(\mu_2 - \frac{\|\sigma_2\|^2}{2}\right)\frac{T}{2} + \sigma_2 W^P\left(\frac{T}{2}\right)} \\
& \left(\mu_1 - \frac{\|\sigma_1\|^2}{2}\right)T + \sigma_1 W^P(T) > \left(\mu_1 - \frac{\|\sigma_1\|^2}{2}\right)\left(\frac{T}{2}\right) + \sigma_1 W^P\left(\frac{T}{2}\right) + \left(\mu_2 - \frac{\|\sigma_2\|^2}{2}\right)\frac{T}{2} + \\
& \sigma_2 W^P\left(\frac{T}{2}\right)
\end{aligned}$$

$$\sigma_1 \left(W^P(T) - W^P\left(\frac{T}{2}\right) \right) - \sigma_2 W^P\left(\frac{T}{2}\right) > \left(\mu_2 - \frac{\|\sigma_2\|^2}{2} - \mu_1 + \frac{\|\sigma_1\|^2}{2} \right) \left(\frac{T}{2} \right)$$

$$\sigma_1 \left(W^P(T) - W^P\left(\frac{T}{2}\right) \right) - \sigma_2 W^P\left(\frac{T}{2}\right) > \left(\mu_2 - \frac{v_2^2}{2} - \mu_1 + \frac{v_1^2}{2} \right) \left(\frac{T}{2} \right)$$

with particular reference with respect to the *historical probability measure* $\mathbb{P}$. Since

$$\sigma_1 \left(W^P(T) - W^P\left(\frac{T}{2}\right) \right) \stackrel{\mathbb{P}}{\sim} N \left(0, \|\sigma_1\|^2 \frac{T}{2} \right)$$

is *independent* of

$$-\sigma_2 W^P\left(\frac{T}{2}\right) \stackrel{\mathbb{P}}{\sim} N \left(0, \|\sigma_2\|^2 \frac{T}{2} \right),$$

w.r.t. the historical probability measure, the $\mathbb{P}$ -distribution of their sum

$$\sigma_1 \left(W^P(T) - W^P\left(\frac{T}{2}\right) \right) - \sigma_2 W^P\left(\frac{T}{2}\right)$$

is $N\left(0, \|\sigma_1\|^2 \frac{T}{2} + \|\sigma_2\|^2 \frac{T}{2}\right)$. Hence we compute

$$\begin{aligned}
& \mathbb{P}\left(N\left(0, \|\sigma_1\|^2 \frac{T}{2} + \|\sigma_2\|^2 \frac{T}{2}\right) > \left(\mu_2 - \frac{v_2^2}{2} - \mu_1 + \frac{v_1^2}{2}\right) \left(\frac{T}{2}\right)\right) = \\
& = \mathbb{P}\left(N\left(0, \frac{T}{2} (\|\sigma_1\|^2 + \|\sigma_2\|^2)\right) > \left(\mu_2 - \frac{v_2^2}{2} - \mu_1 + \frac{v_1^2}{2}\right) \left(\frac{T}{2}\right)\right) \\
& = \mathbb{P}\left(\sqrt{\frac{T}{2} (\|\sigma_1\|^2 + \|\sigma_2\|^2)} Z > \left(\mu_2 - \frac{v_2^2}{2} - \mu_1 + \frac{v_1^2}{2}\right) \left(\frac{T}{2}\right)\right) \\
& = \mathbb{P}\left(Z > \frac{\left(\mu_2 - \mu_1 + \frac{v_1^2 - v_2^2}{2}\right) \sqrt{\frac{T}{2}}}{\sqrt{\|\sigma_1\|^2 + \|\sigma_2\|^2}}\right) \text{ where } Z \stackrel{\mathbb{P}}{\sim} N(0, 1) \\
& = \mathbb{P}\left(Z < -\frac{\left(\mu_2 - \mu_1 + \frac{v_1^2 - v_2^2}{2}\right) \sqrt{\frac{T}{2}}}{\sqrt{v_1^2 + v_2^2}}\right) \\
& = N\left(Z < -\frac{\left(\mu_2 - \mu_1 + \frac{v_1^2 - v_2^2}{2}\right) \sqrt{\frac{T}{2}}}{\sqrt{v_1^2 + v_2^2}}\right)
\end{aligned}$$

Similarly, $\mathbb{Q}\left(S_1(T) > S_1\left(\frac{T}{2}\right) \cdot S_2\left(\frac{T}{2}\right)\right) =$

$$= \mathbb{Q}\left(e^{\left(r - \frac{\|\sigma_1\|^2}{2}\right)T + \sigma_1 W^Q(T)} > e^{\left(r - \frac{\|\sigma_1\|^2}{2}\right)\frac{T}{2} + \sigma_1 W^Q\left(\frac{T}{2}\right)} \cdot e^{\left(r - \frac{\|\sigma_2\|^2}{2}\right)\frac{T}{2} + \sigma_2 W^Q\left(\frac{T}{2}\right)}\right),$$

where W^Q is a 2-dimensional standard Brownian motion with respect

to the *risk-neutral probability* $\mathbb{Q}$. Therefore

$$\begin{aligned}
& \mathbb{Q} \left(S_1(T) > S_1 \left(\frac{T}{2} \right) \cdot S_2 \left(\frac{T}{2} \right) \right) = \\
&= \mathbb{Q} \left(\sigma_1 \left(W^Q(T) - W^Q \left(\frac{T}{2} \right) \right) - \sigma_2 W^Q \left(\frac{T}{2} \right) > \left(r - \frac{v_2^2}{2} - r + \frac{v_1^2}{2} \right) \left(\frac{T}{2} \right) \right) \\
&= \mathbb{Q} \left(\sigma_1 \left(W^Q(T) - W^Q \left(\frac{T}{2} \right) \right) - \sigma_2 W^Q \left(\frac{T}{2} \right) > \left(\frac{v_1^2 - v_2^2}{2} \right) \frac{T}{2} \right) \\
&= \mathbb{Q} \left(\|\sigma_1\| \sqrt{\frac{T}{2}} Z_1^Q + \|\sigma_2\| \sqrt{\frac{T}{2}} Z_2^Q > \left(\frac{v_1^2 - v_2^2}{2} \right) \frac{T}{2} \right) \text{ where } Z_1^Q, Z_2^Q \stackrel{\mathbb{Q}}{\sim} N(0, 1) \text{ and indep.} \\
&= \mathbb{Q} \left(\sqrt{\|\sigma_1\|^2 + \|\sigma_2\|^2} \sqrt{\frac{T}{2}} Z^Q > \left(\frac{v_1^2 - v_2^2}{2} \right) \frac{T}{2} \right) \text{ where } Z^Q \stackrel{\mathbb{Q}}{\sim} N(0, 1) \\
&= \mathbb{Q} \left(\sqrt{v_1^2 + v_2^2} \sqrt{\frac{T}{2}} Z^Q > \left(\frac{v_1^2 - v_2^2}{2} \right) \frac{T}{2} \right) \\
&= \mathbb{Q} \left(Z^Q > \frac{\left(\frac{v_1^2 - v_2^2}{2} \right) \sqrt{\frac{T}{2}}}{\sqrt{v_1^2 + v_2^2}} \right) = N \left(-\frac{\left(\frac{v_1^2 - v_2^2}{2} \right) \sqrt{\frac{T}{2}}}{\sqrt{v_1^2 + v_2^2}} \right)
\end{aligned}$$

2. **(8 points)** A generic random variable X has cumulative distribution function F_X . Explain how to sample X with distribution function F_X by means of the inverse function method.

Consider now the random variable X whose cumulative distribution function is

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x^3 & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1 \end{cases}$$

Apply the inverse transform method to sample X from a uniform random variable.

Solution: The function F_X is strictly increasing for $0 \leq x \leq 1$, and therefore can be inverted. Let $u \in (0; 1)$ then

$$\begin{aligned}
F_X(x) &= u \\
x^3 &= u \\
x &= \sqrt[3]{u} \\
F_X^{-1}(u) &= \sqrt[3]{u}
\end{aligned}$$

Therefore if $U \sim \text{Uniform}(0;1)$ by the inverse transform method $F_X^{-1}(U) = \sqrt[3]{U} \sim F_X$

3. **(6 points)** Explain how the antithetic variate technique works in reducing the variance of the Monte Carlo estimate. When does this technique reduce the variance effectively?
4. **(8 points)** State and derive the formula for the *American perpetual put option price* within the framework of the Black-Scholes market.

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

February 2013

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(12 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Suppose that a European financial derivative on S_1 and S_2 with maturity T trades at time $t \in [0; T]$ at the price

$$F(t, S_1(t), S_2(t)) = e^{\alpha(T-t)} \cdot (S_1(t))^{\frac{1}{4}} \cdot S_2(t),$$

for $\alpha \in \mathfrak{R}$. Find the value of $\alpha \in \mathfrak{R}$ such that $F(t, S_1(t), S_2(t))$ is *the no-arbitrage price* for the derivative in the market (*Your final result must be written in terms of the parameters ρ, v_1, v_2 and r*). **(8 points)**.

Compute the sensitivity of the initial no-arbitrage price of the derivative $F(0, S_1(0), S_2(0))$ with respect to v_1 , the volatility norm of S_1 . Analyze the behavior of $F(0, S_1(0), S_2(0))$ as v_1 varies with respect to the other volatility parameters ρ and v_2 **(4 points)**.

Solution: $F(t, S_1(t), S_2(t))$ is a no-arbitrage price for the European derivative on S_1 and S_2 with maturity T if and only if the function F satisfies the Black-Scholes PDE:

$$\begin{aligned} \frac{\partial F(t; S_1, S_2)}{\partial t} + \sum_{i=1}^2 \frac{\partial F(t; S_1, S_2)}{\partial S_i} \cdot S_i r + \frac{1}{2} \sum_{i,j=1}^2 \frac{\partial^2 F(t; S_1, S_2)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j = \\ = F(t; S_1, S_2) r \text{ for any } S_1, S_2 > 0 \text{ and for any } t \in (0; T) \end{aligned}$$

Therefore, we first compute the derivatives of

$$F(t; S_1, S_2) = e^{\alpha(T-t)} \cdot (S_1)^{\frac{1}{4}} \cdot S_2$$

as follows:

$$\begin{aligned} \frac{\partial F(t; S_1, S_2)}{\partial t} &= -\alpha \cdot e^{\alpha(T-t)} \cdot (S_1)^{\frac{1}{4}} \cdot S_2 = -\alpha F(t; S_1, S_2) \\ \frac{\partial F(t; S_1, S_2)}{\partial S_1} &= \frac{1}{4} e^{\alpha(T-t)} \cdot (S_1)^{-\frac{3}{4}} \cdot S_2 \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1^2} = \frac{-3}{16} e^{\alpha(T-t)} \cdot (S_1)^{-\frac{7}{4}} \cdot S_2 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_2} &= e^{\alpha(T-t)} \cdot (S_1)^{\frac{1}{4}} \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_2^2} = 0 \\ \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1 \partial S_2} &= \frac{1}{4} e^{\alpha(T-t)} \cdot (S_1)^{-\frac{3}{4}}. \end{aligned}$$

We then plug the derivatives into the above PDE

$$\begin{aligned} \underbrace{-\alpha \cdot e^{\alpha(T-t)} \cdot (S_1)^{\frac{1}{4}} \cdot S_2}_{\frac{\partial F(t; S_1, S_2)}{\partial t}} + \underbrace{\frac{1}{4} e^{\alpha(T-t)} \cdot (S_1)^{-\frac{3}{4}} \cdot S_2 \cdot S_1 r}_{\frac{\partial F(t; S_1, S_2)}{\partial S_1}} + \underbrace{e^{\alpha(T-t)} \cdot (S_1)^{\frac{1}{4}} \cdot S_2 r}_{\frac{\partial F(t; S_1, S_2)}{\partial S_2}} + \\ + \frac{1}{2} \underbrace{\frac{-3}{16} e^{\alpha(T-t)} \cdot (S_1)^{-\frac{7}{4}} \cdot S_2}_{\frac{\partial^2 F(t; S_1, S_2)}{\partial S_1^2}} \|\sigma_1\|^2 S_1^2 + 0 + \underbrace{\frac{1}{4} e^{\alpha(T-t)} \cdot (S_1)^{-\frac{3}{4}} \sigma_1^T \sigma_2 S_1 S_2}_{\frac{\partial^2 F(t; S_1, S_2)}{\partial S_1 \partial S_2}} \\ = e^{\alpha(T-t)} \cdot (S_1)^{\frac{1}{4}} \cdot S_2 \cdot r \text{ for any } S_1, S_2 > 0 \text{ and for any } t \in (0; T) \end{aligned}$$

This equation simplifies to

$$-\alpha + \frac{1}{4} r + r + \frac{1}{2} \left(\frac{-3}{16} \right) \|\sigma_1\|^2 + \frac{1}{4} \sigma_1^T \sigma_2 = r$$

delivering

$$\begin{aligned}\alpha &= \frac{1}{4}r + \frac{1}{2} \left(\frac{-3}{16} \right) \|\sigma_1\|^2 + \frac{1}{4}\sigma_1^T \sigma_2 \\ &= \frac{1}{4}r - \frac{3}{32}v_1^2 + \frac{1}{4}\rho v_1 v_2.\end{aligned}$$

The sensitivity of the initial no-arbitrage price of the derivative $F(0, S_1(0), S_2(0)) = e^{\alpha T} \cdot (S_1(0))^{\frac{1}{4}} \cdot S_2(0)$, where $\alpha = \frac{1}{4}r - \frac{3}{32}v_1^2 + \frac{1}{4}\rho v_1 v_2$, with respect to v_1 is

$$\begin{aligned}\frac{\partial F(0; S_1(0), S_2(0))}{\partial v_1} &= \frac{\partial}{\partial v_1} \left(e^{\alpha T} \cdot (S_1(0))^{\frac{1}{4}} \cdot S_2(0) \right) \\ &= (S_1(0))^{\frac{1}{4}} \cdot S_2(0) \cdot \frac{\partial}{\partial v_1} (e^{\alpha T}) \\ &= (S_1(0))^{\frac{1}{4}} \cdot S_2(0) \cdot T \cdot e^{\alpha T} \frac{\partial}{\partial v_1} (\alpha) \\ &= (S_1(0))^{\frac{1}{4}} \cdot S_2(0) \cdot T \cdot e^{\alpha T} \frac{\partial}{\partial v_1} \left(\frac{1}{4}r - \frac{3}{32}v_1^2 + \frac{1}{4}\rho v_1 v_2 \right) \\ &= (S_1(0))^{\frac{1}{4}} \cdot S_2(0) \cdot T \cdot e^{\alpha T} \left(-\frac{6}{32}v_1 + \frac{1}{4}\rho v_2 \right)\end{aligned}$$

Hence the sensitivity is positive if and only if

$$\begin{aligned}-\frac{6}{32}v_1 + \frac{1}{4}\rho v_2 &> 0 \\ v_1 &< \frac{4}{3}\rho v_2\end{aligned}$$

Therefore $F(0; S_1(0), S_2(0))$ is an increasing function of v_1 if $v_1 < \frac{4}{3}\rho v_2$. On the contrary, if $v_1 \geq \frac{4}{3}\rho v_2$, the function $F(0; S_1(0), S_2(0))$ is decreasing with respect to v_1 .

2. **(8 points)** Explain how the control variate technique works in reducing the variance of a Monte Carlo estimate.
3. **(4 points)** State the backward recursive formula to price a finite-maturity American put option in a discrete-time setting.
4. **(7 points)** State Ito formula in the jump diffusion setting (specify all the assumptions on the involved stochastic processes).

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

September 2013

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(12 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the stochastic differential of the process

$$Y(t) = (S_1(t))^\alpha \cdot S_2(t),$$

with respect to the historical and the risk neutral probability measure (*Your final result must be written in terms of the parameters ρ, v_1, v_2 and r*). **(8 points)**.

Suppose that a European financial derivative on S_1 and S_2 with maturity T has terminal payoff

$$X(T) = Y(T) = (S_1(T))^\alpha \cdot S_2(T),$$

for $\alpha \in \mathbb{R}$. Assume that $v_1 - 2\rho v_2 \neq 0$. Find the value of $\alpha \in \mathbb{R}$ such that $S_X(0)$, the initial no-arbitrage price for the derivative in the market, is equal to $e^{\alpha r T}$ and not equal to 1, that is

$$S_X(0) = e^{\alpha r T} \neq 1$$

(4 points).

Solution: To compute the stochastic differential of the process

$$Y(t) = F(t; S_1(t), S_2(t)) = (S_1(t))^\alpha \cdot S_2(t),$$

we first compute the derivatives of

$$F(t; S_1, S_2) = (S_1)^\alpha \cdot S_2.$$

We have

$$\begin{aligned} \frac{\partial F(t; S_1, S_2)}{\partial t} &= 0 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_1} &= \alpha (S_1)^{\alpha-1} \cdot S_2 \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1^2} = \alpha(\alpha-1) (S_1)^{\alpha-2} \cdot S_2 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_2} &= (S_1)^\alpha \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_2^2} = 0 \\ \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1 \partial S_2} &= \alpha (S_1)^{\alpha-1}. \end{aligned}$$

Ito formula delivers the stochastic differential of Y as follows

$$\begin{aligned} dY &= 0dt + \alpha (S_1)^{\alpha-1} \cdot S_2 dS_1 + (S_1)^\alpha dS_2 + \frac{1}{2} \alpha(\alpha-1) (S_1)^{\alpha-2} \cdot S_2 \|\sigma_1\|^2 S_1^2 dt + \\ &\quad + 0 + \alpha (S_1)^{\alpha-1} \sigma_1 \cdot \sigma_2 S_1 S_2 dt + 0 \\ &= (S_1)^\alpha \cdot S_2 \cdot \left\{ \left(\alpha \mu_1 + \mu_2 + \frac{1}{2} \alpha(\alpha-1) \|\sigma_1\|^2 + \alpha \sigma_1 \cdot \sigma_2 \right) dt + (\alpha \sigma_1 + \sigma_2) \cdot dW^P(t) \right\} \\ &= Y \left\{ \left(\alpha \mu_1 + \mu_2 + \frac{1}{2} \alpha(\alpha-1) v_1^2 + \alpha \rho v_1 v_2 \right) dt + [\alpha v_1 + \rho v_2, \sqrt{1-\rho^2} v_2] dW^P(t) \right\} \\ &= Y \left\{ \left(\alpha r + r + \frac{1}{2} \alpha(\alpha-1) v_1^2 + \alpha \rho v_1 v_2 \right) dt + [\alpha v_1 + \rho v_2, \sqrt{1-\rho^2} v_2] dW^Q(t) \right\} \\ &= Y \left\{ \left((\alpha+1)r + \frac{1}{2} \alpha(\alpha-1) v_1^2 + \alpha \rho v_1 v_2 \right) dt + [\alpha v_1 + \rho v_2, \sqrt{1-\rho^2} v_2] dW^Q(t) \right\} \end{aligned}$$

where $[\alpha v_1 + \rho v_2, \sqrt{1-\rho^2} v_2] dW^Q(t) \sim \left\| [\alpha v_1 + \rho v_2, \sqrt{1-\rho^2} v_2] \right\| dB(t)$
 where B is a standard one dimensional Q -Brownian motion.

The process Y is lognormal with respect to both the *risk neutral* and the *historical* probability measure, with volatility σ_Y given by

$$\sigma_Y^2 = \left\| [\alpha v_1 + \rho v_2, \sqrt{1-\rho^2} v_2] \right\|^2 = (\alpha v_1 + \rho v_2)^2 + \left(\sqrt{1-\rho^2} v_2 \right)^2 = \alpha^2 v_1^2 + 2\alpha \rho v_1 v_2 + v_2^2.$$

The risk-neutral drift is

$$\mu_Y^Q = (\alpha + 1)r + \frac{1}{2}\alpha(\alpha - 1)v_1^2 + \alpha\rho v_1 v_2$$

The no-arbitrage price of the derivative X at the initial date is

$$\begin{aligned} S_X(0) &= \mathbb{E}^Q [X(T) e^{-rT}] = \mathbb{E}^Q [Y(T) e^{-rT}] \\ &= Y(0) e^{\mu_Y^Q T} e^{-rT} \quad \text{because } Y \text{ is lognormal} \\ &= 1 \exp \left(\left((\alpha + 1)r + \frac{1}{2}\alpha(\alpha - 1)v_1^2 + \alpha\rho v_1 v_2 \right) T \right) \exp(-rT) \\ &= \exp \left(\left(\alpha r + \frac{1}{2}\alpha(\alpha - 1)v_1^2 + \alpha\rho v_1 v_2 \right) T \right). \end{aligned}$$

The initial no-arbitrage price satisfies the constraint

$$S_X(0) = e^{\alpha r T}$$

if and only if

$$\begin{aligned} \left(\alpha r + \frac{1}{2}\alpha(\alpha - 1)v_1^2 + \alpha\rho v_1 v_2 \right) T &= \alpha r T \\ \frac{1}{2}\alpha(\alpha - 1)v_1^2 + \alpha\rho v_1 v_2 &= 0 \\ \frac{1}{2}\alpha v_1(\alpha v_1 - v_1 + 2\rho v_2) &= 0 \end{aligned}$$

the last equation delivers the values

$$\alpha = 0 \text{ and } \alpha = \frac{v_1 - 2\rho v_2}{v_1} \neq 0.$$

Since $\alpha = 0$ implies $S_X(0) = e^{\alpha r T} = 1$, the solution is $\alpha = \frac{v_1 - 2\rho v_2}{v_1} \neq 0$, that implies $S_X(0) \neq 1$.

2. **(8 points)** Derive the *pathwise derivative estimate* for the Delta and the Vega of a call option in the Black-Scholes model. Does this method provide always an *unbiased* estimate of the greeks of a derivative? Can it always be applied?
3. **(7 points)** Explain how the antithetic variate technique works in reducing the variance of a Monte Carlo estimate. When is this technique effective to reduce the variance? Does it provide an *unbiased* estimate?.

4. **(4 points)** In Merton's jump-diffusion model, write the stochastic differential equation satisfied by the risky security S with respect to the *historical probability measure* $\mathbf{P}$ (specify all the assumptions on the involved stochastic processes).

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

January 2014

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(8 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the stochastic differential of the process

$$Y(t) = S_1(t) \cdot \sqrt{S_2(t)},$$

with respect to the historical and the risk neutral probability measure (*Your final result must be written in terms of the parameters ρ, v_1, v_2 and r .*

2. **(4 points)** Suppose that a European financial derivative with maturity T has terminal payoff

$$X(T) = \begin{cases} 1 & \text{if } Y(T) \geq Y\left(\frac{T}{2}\right) \\ 0 & \text{otherwise} \end{cases}$$

where Y is defined in the previous point. Find the *initial no-arbitrage price* $S_X(0)$ for the derivative in the market (*Your final result must be written in terms of the parameters ρ, v_1, v_2 and r*).

3. **(4 points)** In the discrete-time framework state the backward recursive formula to compute the price of an American *digital* option on the risky asset S with maturity T whose *immediate payoff at t* is $X(t) = 1$ if $S(t) < K$, and 0 otherwise.
4. **(7 points)** Derive the *pathwise derivative estimate* for the Delta and the Vega of a call option in the Black-Scholes model. Does this method provide always an *unbiased* estimate of the greeks of a derivative? Can it always be applied?
5. **(8 points)** State and derive the formula for the *American perpetual put option price* within the framework of the Black-Scholes market.

Solution of the exercises

1. To compute the stochastic differential of the process

$$Y(t) = F(t; S_1(t), S_2(t)) = S_1(t) \cdot \sqrt{S_2(t)},$$

we first compute the derivatives of

$$F(t; S_1, S_2) = S_1 \cdot \sqrt{S_2}.$$

We have

$$\begin{aligned} \frac{\partial F(t; S_1, S_2)}{\partial t} &= 0 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_1} &= \sqrt{S_2} \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1^2} = 0 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_2} &= S_1 \cdot \frac{1}{2} \frac{1}{\sqrt{S_2}} \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_2^2} = -S_1 \cdot \frac{1}{4} \frac{1}{\sqrt{(S_2)^3}} \\ \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1 \partial S_2} &= \frac{1}{2} \frac{1}{\sqrt{S_2}}. \end{aligned}$$

Ito formula delivers the stochastic differential of Y as follows

$$\begin{aligned} dY &= 0dt + \sqrt{S_2} dS_1 + S_1 \cdot \frac{1}{2} \frac{1}{\sqrt{S_2}} dS_2 + 0 + \\ &\quad + \frac{1}{2} \left(-S_1 \cdot \frac{1}{4} \frac{1}{\sqrt{(S_2)^3}} \right) \cdot \|\sigma_2\|^2 S_2^2 dt + \frac{1}{2} \frac{1}{\sqrt{S_2}} \sigma_1 \cdot \sigma_2 S_1 S_2 dt + 0 \\ &= S_1 \cdot \sqrt{S_2} \left\{ \left(\mu_1 + \frac{1}{2} \mu_2 - \frac{1}{8} \|\sigma_2\|^2 + \frac{1}{2} \sigma_1 \cdot \sigma_2 \right) dt + \left(\sigma_1 + \frac{1}{2} \sigma_2 \right) \cdot dW^P(t) \right\} \\ &= Y \left\{ \left(\mu_1 + \frac{1}{2} \mu_2 - \frac{1}{8} v_2^2 + \frac{1}{2} \rho v_1 v_2 \right) dt + \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^P(t) \right\} \\ &= Y \left\{ \left(r + \frac{1}{2} r - \frac{1}{8} v_2^2 + \frac{1}{2} \rho v_1 v_2 \right) dt + \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\} \\ &= Y \left\{ \left(\frac{3}{2} r - \frac{1}{8} v_2^2 + \frac{1}{2} \rho v_1 v_2 \right) dt + \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\} \end{aligned}$$

where $\left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \sim \left\| \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] \right\| dB(t)$
 where B^Q is a standard one dimensional Q -Brownian motion.

The process Y is lognormal with respect to both the *risk neutral* and the *historical* probability measure, with volatility σ_Y given by

$$\begin{aligned}\sigma_Y^2 &= \left\| \left[v_1 + \frac{1}{2}\rho v_2, \frac{1}{2}\sqrt{1-\rho^2}v_2 \right] \right\|^2 = \left(v_1 + \frac{1}{2}\rho v_2 \right)^2 + \left(\frac{1}{2}\sqrt{1-\rho^2}v_2 \right)^2 \\ &= v_1^2 + \rho v_1 v_2 + \frac{1}{4}v_2^2.\end{aligned}$$

The risk-neutral drift is

$$\mu_Y^Q = \frac{3}{2}r - \frac{1}{8}v_2^2 + \frac{1}{2}\rho v_1 v_2$$

2. The no-arbitrage price of the derivative X at the initial date is

$$\begin{aligned}S_X(0) &= \mathbb{E}^Q [X(T) e^{-rT}] = e^{-rT} \mathbb{Q} \left[Y(T) \geq Y\left(\frac{T}{2}\right) \right] \\ &= e^{-rT} \mathbb{Q} \left[\frac{Y(T)}{Y\left(\frac{T}{2}\right)} \geq 1 \right] = e^{-rT} \mathbb{Q} \left[e^{\left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right)\frac{T}{2} + \sigma_Y(B^Q(T) - B^Q(\frac{T}{2}))} \geq 1 \right] \\ &= e^{-rT} \mathbb{Q} \left[\left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right)\frac{T}{2} + \sigma_Y \left(B^Q(T) - B^Q\left(\frac{T}{2}\right) \right) \geq 0 \right] \\ &= e^{-rT} \mathbb{Q} \left[\left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right)\frac{T}{2} + \sigma_Y \sqrt{\frac{T}{2}} Z \geq 0 \right] \quad \text{where } Z \stackrel{\mathbb{Q}}{\sim} N(0, 1) \\ &= e^{-rT} \mathbb{Q} \left[Z \geq -\left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right) \sqrt{\frac{T}{2}} \frac{1}{\sigma_Y} \right] \\ &= e^{-rT} \mathbb{Q} \left[Z \leq \left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right) \sqrt{\frac{T}{2}} \frac{1}{\sigma_Y} \right] \\ &= e^{-rT} N \left[\left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right) \sqrt{\frac{T}{2}} \frac{1}{\sigma_Y} \right]\end{aligned}$$

and since $\left(\mu_Y^Q - \frac{\sigma_Y^2}{2}\right) \sqrt{\frac{T}{2}} \frac{1}{\sigma_Y} = \sqrt{\frac{T}{2}} \frac{1}{\sigma_Y} \left(\frac{3}{2}r - \frac{1}{8}v_2^2 + \frac{1}{2}\rho v_1 v_2 - \frac{1}{2}(v_1^2 + \rho v_1 v_2 + \frac{1}{4}v_2^2)\right)$
 $= \sqrt{\frac{T}{2}} \frac{1}{\sigma_Y} \left(-\frac{1}{2}v_1^2 - \frac{1}{4}v_2^2 + \frac{3}{2}r\right)$, we get

$$S_X(0) = e^{-rT} N \left[\left(-\frac{1}{2}v_1^2 - \frac{1}{4}v_2^2 + \frac{3}{2}r\right) \sqrt{\frac{T}{2}} \frac{1}{\sqrt{v_1^2 + \rho v_1 v_2 + \frac{1}{4}v_2^2}} \right]$$

3. The *immediate payoff* at $t \in \{0, \dots, T\}$ of the American digital option is

$$X(t) = \mathbb{I}_{S(t) < K} = \begin{cases} 1 & \text{if } S(t) < K \\ 0 & \text{otherwise} \end{cases}$$

where $\mathbb{I}_{S(t) < K}$ denotes the indicator function of the event $\{S(t) < K\}$. The discrete backward recursive formula to compute the price of the American *digital* option is

$$V(T) = X(T) = \mathbb{I}_{S(T) < K} = \begin{cases} 1 & \text{if } S(T) < K \\ 0 & \text{otherwise} \end{cases}$$

and

$$V(t) = \max \left\{ \mathbb{I}_{S(t) < K}; \mathbf{E}^Q \left[\frac{V(t+1)}{1+r(t)} \middle| \mathcal{P}_t \right] \right\}$$

for $t = T-1, \dots, 0$.

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

February 2014

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(7 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the stochastic differential of

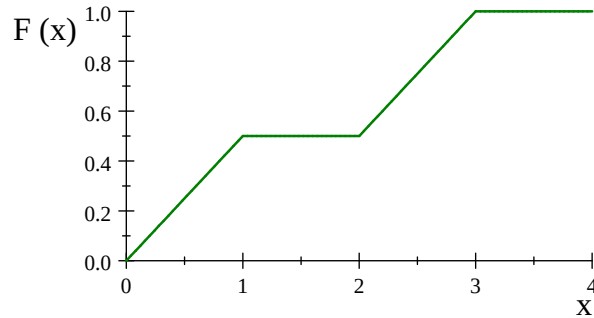
$$Y(t) = \ln \left(S_1(t) \cdot \sqrt{S_2(t)} \right)$$

with respect to both the *historical probability measure* $\mathbb{P}$ and the *risk-neutral probability measure* $\mathbb{Q}$. Which is the distribution of the process Y with respect to both probabilities? (*Your final result must be written in terms of the parameters $\mu_1, \mu_2, r, \rho, v_1, v_2$.*)

2. **(11 points)** A generic random variable X has cumulative distribution function F_X . Explain how to sample X with distribution function F_X by means of the inverse function method. **(8 points)**

Consider now the random variable X whose cumulative distribution function is

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{2}x & \text{if } 0 \leq x < 1 \\ \frac{1}{2} & \text{if } 1 \leq x < 2 \\ \frac{1}{2}x - \frac{1}{2} & \text{if } 2 \leq x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$



Graph of the function $F_X(x)$

Apply the inverse transform method to sample X from a uniform random variable **(3 points)**.

3. **(6 points)** Define the central-difference estimator for the sensitivities of a given derivative. Is this estimator unbiased? Justify your answer.
4. **(7 points)** State Ito formula in the jump diffusion setting (specify all the assumptions on the involved stochastic processes).

Solution of the exercises

1. We compute the stochastic differential of the process

$$Y(t) = \ln \left(S_1(t) \cdot \sqrt{S_2(t)} \right),$$

via Ito formula. We first compute the derivatives of

$$\begin{aligned} F(t; S_1, S_2) &= \ln \left(S_1 \cdot \sqrt{S_2} \right) = \\ &= \ln(S_1) + \frac{1}{2} \ln(S_2) \end{aligned}$$

We have

$$\begin{aligned} \frac{\partial F(t; S_1, S_2)}{\partial t} &= 0 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_1} &= \frac{1}{S_1} \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1^2} = -\frac{1}{S_1^2} \\ \frac{\partial F(t; S_1, S_2)}{\partial S_2} &= \frac{1}{2} \frac{1}{S_2} \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_2^2} = -\frac{1}{2} \frac{1}{S_2^2} \\ \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1 \partial S_2} &= 0. \end{aligned}$$

Ito formula delivers the stochastic differential of Y as follows

$$\begin{aligned} dY &= 0dt + \frac{1}{S_1} dS_1 + \frac{1}{2} \frac{1}{S_2} dS_2 + \\ &\quad + \frac{1}{2} \left(-\frac{1}{S_1^2} \right) \cdot \|\sigma_1\|^2 S_1^2 dt + \frac{1}{2} \left(-\frac{1}{2} \frac{1}{S_2^2} \right) \cdot \|\sigma_2\|^2 S_2^2 dt + 0 \\ &= \left(\mu_1 + \frac{1}{2} \mu_2 - \frac{1}{2} \|\sigma_1\|^2 - \frac{1}{4} \|\sigma_2\|^2 \right) dt + \left(\sigma_1 + \frac{1}{2} \sigma_2 \right) \cdot dW^P(t) \\ &= \left(\mu_1 + \frac{1}{2} \mu_2 - \frac{1}{2} v_1^2 - \frac{1}{4} v_2^2 \right) dt + \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^P(t) \\ &= \left(r + \frac{1}{2} r - \frac{1}{2} v_1^2 - \frac{1}{4} v_2^2 \right) dt + \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \\ &= \left(\frac{3}{2} r - \frac{1}{2} v_1^2 - \frac{1}{4} v_2^2 \right) dt + \left[v_1 + \frac{1}{2} \rho v_2, \frac{1}{2} \sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \end{aligned}$$

where $\left[v_1 + \frac{1}{2}\rho v_2, \frac{1}{2}\sqrt{1-\rho^2}v_2 \right] dW^Q(t) \sim \left\| \left[v_1 + \frac{1}{2}\rho v_2, \frac{1}{2}\sqrt{1-\rho^2}v_2 \right] \right\| dB^Q(t)$
where B^Q is a standard one dimensional Q -Brownian motion.

The process Y is a generalized Brownian motion to both the *risk neutral* and the *historical* probability measure. In particular

$$Y(t) \stackrel{Q}{\sim} \mathcal{N}(\mu_Y^Q \cdot t, \sigma_Y^2 \cdot t)$$

where σ_Y is

$$\begin{aligned} \sigma_Y^2 &= \left\| \left[v_1 + \frac{1}{2}\rho v_2, \frac{1}{2}\sqrt{1-\rho^2}v_2 \right] \right\|^2 = \left(v_1 + \frac{1}{2}\rho v_2 \right)^2 + \left(\frac{1}{2}\sqrt{1-\rho^2}v_2 \right)^2 \\ &= v_1^2 + \rho v_1 v_2 + \frac{1}{4}v_2^2. \end{aligned}$$

The parameter μ_Y^Q is

$$\mu_Y^Q = \frac{3}{2}r - \frac{1}{2}v_1^2 - \frac{1}{4}v_2^2.$$

The parameters with respect to the historical probability measure P are

$$Y(t) \stackrel{P}{\sim} \mathcal{N}(\mu_Y^P \cdot t, \sigma_Y^2 \cdot t)$$

where μ_Y^P is

$$\mu_Y^P = \mu_1 + \frac{1}{2}\mu_2 - \frac{1}{2}v_1^2 - \frac{1}{4}v_2^2.$$

2. From the graph of the function F_X we see that

$$F_X(x) = u$$

has one and only one solution $x \geq 0$ if $u \in (0, \frac{1}{2})$ or $u \in (\frac{1}{2}, 1)$. In fact, if $u \in (0, \frac{1}{2})$, equation

$$F_X(x) = \frac{1}{2}x = u$$

delivers $x = 2u \in (0, 1)$. If $u \in (\frac{1}{2}, 1)$, equation

$$F_X(x) = \frac{1}{2}x - \frac{1}{2} = u$$

delivers $x = 2u + 1 \in (2, 3)$.

If $u = \frac{1}{2}$, equation

$$F_X(x) = u$$

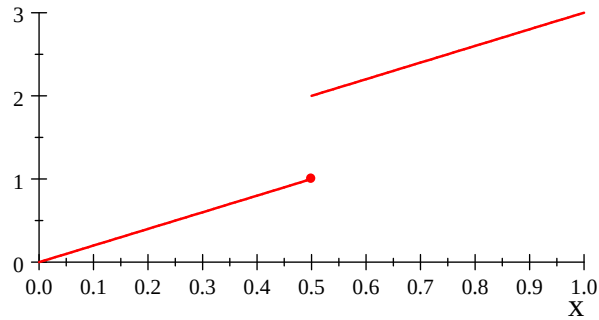
is satisfied for *all* $x \in [1, 2]$. The definition of the generalized inverse function delivers the smallest of these solutions

$$\begin{aligned} F_X^{-1}\left(\frac{1}{2}\right) &= \inf \left\{ t : F_X(t) \geq \frac{1}{2} \right\} \\ &= \inf \{ t : t \in [1, 2] \} = \\ &= 1 \end{aligned}$$

Summing up herefore

$$F_X^{-1}(u) = \begin{cases} 2u & \text{if } u \in [0, \frac{1}{2}] \\ 2u + 1 & \text{if } u \in (\frac{1}{2}, 1] \end{cases}$$

Therefore if $U \sim \text{Uniform}(0; 1)$ by the inverse transform method $F_X^{-1}(U) \sim F_X$



Graph of the function $F_X^{-1}(x)$

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

September 2014

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(8 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Denote with r the riskless interest rate.

Compute the stochastic differential of the process

$$Y(t) = S_1(t) \cdot (S_2(t))^4,$$

with respect to the historical and the risk neutral probability measure (*Your final result must be written in terms of the parameters ρ, v_1, v_2 and r .*

2. **(4 points)** Consider a forward contract on Y , requiring the long position to buy Y at time $t = T$ at a delivery price F_Y specified at time $t = 0$ in such a way that the initial value of the forward contract is zero. Find the delivery price F_Y . (*Your final result must be written in terms of the parameters ρ, v_1, v_2 and r .*) Does F_Y increase or decrease if the correlation ρ increases? Justify your answer.

3. **(4 points)** In the discrete-time framework state the backward recursive formula to compute the price of an American *put* option on the risky asset S with maturity T .
4. **(7 points)** Define the forward-difference and the central-difference estimator for the sensitivities of a given derivative. Are these estimators unbiased? Which one is superior? Justify your answer.
5. **(8 points)** In the multidimensional Black-Scholes model state and derive the Black-Scholes Partial Differential Equation (PDE). Which derivatives satisfy the PDE?

Solution of the exercises

1. To compute the stochastic differential of the process

$$Y(t) = F(t; S_1(t), S_2(t)) = S_1(t) \cdot S_2(t)^4,$$

we first compute the derivatives of

$$F(t; S_1, S_2) = S_1 \cdot S_2^4.$$

We have

$$\begin{aligned} \frac{\partial F(t; S_1, S_2)}{\partial t} &= 0 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_1} &= S_2^4 \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1^2} = 0 \\ \frac{\partial F(t; S_1, S_2)}{\partial S_2} &= S_1 \cdot 4S_2^3 \text{ and } \frac{\partial^2 F(t; S_1, S_2)}{\partial S_2^2} = 4S_1 \cdot 3S_2^2 \\ \frac{\partial^2 F(t; S_1, S_2)}{\partial S_1 \partial S_2} &= 4S_2^3. \end{aligned}$$

Ito formula delivers the stochastic differential of Y as follows

$$\begin{aligned} dY &= 0dt + S_2^4 dS_1 + S_1 \cdot 4S_2^3 dS_2 + 0 + \\ &\quad + \frac{1}{2} (4S_1 \cdot 3S_2^2) \cdot \|\sigma_2\|^2 S_2^2 dt + 4S_2^3 \sigma_1 \cdot \sigma_2 S_1 S_2 dt + 0 \\ &= S_1 S_2^4 \{ (\mu_1 + 4\mu_2 + 6\|\sigma_2\|^2 + 4\sigma_1 \cdot \sigma_2) dt + (\sigma_1 + 4\sigma_2) \cdot dW^P(t) \} \\ &= Y \left\{ (\mu_1 + 4\mu_2 + 6v_2^2 + 4\rho v_1 v_2) dt + \left[v_1 + 4\rho v_2, 4\sqrt{1 - \rho^2} v_2 \right] dW^P(t) \right\} \\ &= Y \left\{ (r + 4r + 6v_2^2 + 4\rho v_1 v_2) dt + \left[v_1 + 4\rho v_2, 4\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\} \\ &= Y \left\{ (5r + 6v_2^2 + 4\rho v_1 v_2) dt + \left[v_1 + 4\rho v_2, 4\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\} \end{aligned}$$

where $\left[v_1 + 4\rho v_2, 4\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \sim \left\| \left[v_1 + 4\rho v_2, 4\sqrt{1 - \rho^2} v_2 \right] \right\| dB(t)$
where B^Q is a standard one dimensional Q -Brownian motion.

The process Y is lognormal with respect to both the *risk neutral* and the *historical* probability measure, with volatility σ_Y given by

$$\begin{aligned} \sigma_Y^2 &= \left\| \left[v_1 + 4\rho v_2, 4\sqrt{1 - \rho^2} v_2 \right] \right\|^2 = (v_1 + 4\rho v_2)^2 + \left(4\sqrt{1 - \rho^2} v_2 \right)^2 \\ &= v_1^2 + 8\rho v_1 v_2 + 16v_2^2. \end{aligned}$$

The risk-neutral drift is

$$\mu_Y^Q = 5r + 6v_2^2 + 4\rho v_1 v_2$$

2. The delivery price F_Y is settled at $t = 0$ such that

$$\mathbb{E}^Q [(Y(T) - F_Y) e^{-rT}] = 0,$$

that implies

$$F_Y = \mathbb{E}^Q [Y(T)].$$

Since Y is lognormal, we have that

$$\begin{aligned} F_Y &= \mathbb{E}^Q [Y(T)] = Y(0) e^{\mu_Y^Q T} \\ &= 1 \cdot e^{\mu_Y^Q T} \end{aligned}$$

where $\mu_Y^Q = 5r + 6v_2^2 + 4\rho v_1 v_2$. The delivery price is increasing with respect to ρ since

$$\begin{aligned} \frac{\partial F_Y}{\partial \rho} &= T e^{\mu_Y^Q T} \frac{\partial \mu_Y^Q}{\partial \rho} \\ &= T e^{\mu_Y^Q T} 4v_1 v_2 > 0. \end{aligned}$$

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

January 2015

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(8 points)** Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0) = 1$.

Compute the stochastic differential of

$$Y(t) = e^{\alpha t} \cdot (S_1(t)) \cdot (S_2(t))^2,$$

with $\alpha \in \Re$, with respect to the *risk neutral* and the *historical* probability measure.

Is the process Y lognormal? If your answer is positive, compute the drift and the volatility of Y with respect to the *risk neutral* and the *historical* probability measure. (*Your final results must be written in terms of the parameters $\alpha, \rho, v_1, v_2, \mu_1, \mu_2, r$.*)

2. **(4 points)** Suppose now that the process $Y = \{Y(t)\}_{t \in [0, T]}$ defined in the previous point is the *price process* of a European derivative whose payoff at maturity T is

$$Y(T) = e^{\alpha T} \cdot (S_1(T)) \cdot (S_2(T))^2.$$

Find $\alpha \in \mathfrak{R}$ that preserves no-arbitrage in the market constituted by B , S_1 , S_2 and Y .

3. **(5 points)** In the discrete-time framework state the backward recursive formula to compute the price for an American lookback option, whose instantaneous payoff at t is the difference between the underlying's *running maximum* and a fixed strike price K , whenever positive.

Is the time t price of the American derivative a deterministic function of the price of the underlying asset at time t ?

4. **(7 points)** Define the forward-difference estimator and the central-difference estimator for the sensitivities of a given derivative. Are these estimators unbiased? Which is better? Justify your answer.
5. **(7 points)** State and derive the formula of the price of a *perpetual American put option* in the Black-Scholes setting.

Solution of the Exercises

1. Given the function

$$f(t, S_1, S_2) = e^{\alpha t} \cdot S_1 \cdot S_2^2$$

the derivatives are

$$\begin{aligned} \frac{\partial f}{\partial t} &= \alpha \cdot e^{\alpha t} \cdot S_1 \cdot S_2^2, & \frac{\partial f}{\partial S_1} &= e^{\alpha t} \cdot S_2^2, & \frac{\partial^2 f}{\partial S_1^2} &= 0 \\ \frac{\partial f}{\partial S_2} &= 2 \cdot e^{\alpha t} \cdot S_1 \cdot S_2, & \frac{\partial^2 f}{\partial S_2^2} &= 2 \cdot e^{\alpha t} \cdot S_1, & \frac{\partial^2 f}{\partial S_1 \partial S_2} &= 2 \cdot e^{\alpha t} \end{aligned}$$

and therefore

$$\begin{aligned} dY &= \alpha \cdot e^{\alpha t} \cdot S_1 \cdot S_2^2 dt + e^{\alpha t} \cdot S_2^2 dS_1 + 2 \cdot e^{\alpha t} \cdot S_1 \cdot S_2 dS_2 + 0 + \\ &\quad + \frac{1}{2} (2 \cdot e^{\alpha t} \cdot S_1) \|\sigma_2\|^2 S_2^2 dt + (2 \cdot e^{\alpha t}) \sigma_1 \cdot \sigma_2 S_1 S_2 dt + 0 \\ &= e^{\alpha t} \cdot S_1 \cdot S_2^2 \{ (\alpha + \mu_1 + 2\mu_2 + \|\sigma_2\|^2 + 2\sigma_1 \cdot \sigma_2) dt + (\sigma_1 + 2\sigma_2) \cdot dW^P(t) \} \\ &= Y \left\{ (\alpha + \mu_1 + 2\mu_2 + v_2^2 + 2\rho v_1 v_2) dt + \left[v_1 + 2\rho v_2, 2\sqrt{1 - \rho^2} v_2 \right] dW^P(t) \right\} \\ &= Y \left\{ (\alpha + r + 2r + v_2^2 + 2\rho v_1 v_2) dt + \left[v_1 + 2\rho v_2, 2\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\} \\ &= Y \left\{ (\alpha + 3r + v_2^2 + 2\rho v_1 v_2) dt + \left[v_1 + 2\rho v_2, 2\sqrt{1 - \rho^2} v_2 \right] dW^Q(t) \right\}. \end{aligned}$$

Denoting $\sigma_Y = \left[v_1 + 2\rho v_2, 2\sqrt{1 - \rho^2} v_2 \right]$ we observe that $\sigma_Y dW^Q(t) \sim \|\sigma_Y\| dB(t)$, where B is a standard one dimensional Q -Brownian motion.

Therefore, the process Y is lognormal with respect to both the *risk neutral* and the *historical* probability measure, with volatility $\|\sigma_Y\| =$

$\left\| \left[v_1 + 2\rho v_2, 2\sqrt{1 - \rho^2} v_2 \right] \right\|$ given by

$$\|\sigma_Y\|^2 = \left\| \left[v_1 + 2\rho v_2, 2\sqrt{1 - \rho^2} v_2 \right] \right\|^2 = (v_1 + 2\rho v_2)^2 + \left(2\sqrt{1 - \rho^2} v_2 \right)^2 = v_1^2 + 4\rho v_1 v_2 + 4v_2^2$$

and drift

$$\begin{aligned} \mu_Y^P &= \alpha + \mu_1 + 2\mu_2 + v_2^2 + 2\rho v_1 v_2 \quad \text{with respect to the historical probability } P \\ \mu_Y^Q &= \alpha + 3r + v_2^2 + 2\rho v_1 v_2 \quad \text{with respect to the risk neutral probability } Q. \end{aligned}$$

2. The market constituted by B , S_1 and S_2 of the previous point is complete and arbitrage-free. The European derivative whose payoff at maturity T is

$$Y(T) = e^{\alpha T} \cdot (S_1(T)) \cdot (S_2(T))^2,$$

is attainable. No-arbitrage is preserved if and only if its discounted price

$$Y(t) \cdot e^{-rt}$$

is a risk-neutral martingale, that is if and only if

$$d(Y(t) \cdot e^{-rt}) = 0 \cdot dt + () \cdot dW^Q(t).$$

Since

$$\begin{aligned} d(Y(t) \cdot e^{-rt}) &= -r \cdot e^{-rt} Y(t) \cdot dt + e^{-rt} \cdot dY(t) \\ &= -r \cdot e^{-rt} Y(t) \cdot dt + e^{-rt} \cdot Y(t) \cdot \left(\mu_Y^Q \cdot dt + \sigma_Y \cdot dW^Q(t) \right) \\ &= Y(t) \cdot e^{-rt} \cdot \left(\left(\mu_Y^Q - r \right) \cdot dt + \sigma_Y \cdot dW^Q(t) \right) \end{aligned}$$

we obtain that

$$\mu_Y^Q - r = 0 \text{ that is } \mu_Y^Q = r.$$

This yields

$$\alpha + 3r + v_2^2 + 2\rho v_1 v_2 = r,$$

and therefore

$$\alpha = -2r - v_2^2 - 2\rho v_1 v_2.$$

One can reach the same conclusion by applying to the deterministic function

$$F(t, S_1, S_2) = e^{\alpha t} \cdot S_1 \cdot S_2^2$$

the Black-Scholes Partial Differential Equation.

Note that the unique value of α determines a unique terminal payoff at maturity T , and hence a unique derivative as well.

3. Let $F(t) = \max_{0 \leq s \leq t} S(s)$. The immediate payoff of the American option is

$$X(t) = (F(t) - K)^+.$$

Then the discounted value of the American option is given by $\tilde{V}(T) = \tilde{X}(T)$ at $t = T$, and

$$\tilde{V}(t) = \max \left\{ \tilde{X}(t); \mathbf{E}^Q \left[\tilde{V}(t+1) \middle| \mathcal{P}_t \right] \right\}$$

for $t = T - 1, \dots, 0$.

The time t price of the American derivative is not a deterministic function of the price of the underlying asset at time t only, but is a deterministic function of the pair $(S(t), F(t))$. The pricing problem is Markovian with respect to the pair $(S(t), F(t))$.

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

February 2015

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

1. **(4 points)** In the discrete-time framework state the backward recursive formula to compute the price for an American *put* option on a risky asset S .
2. **(7 points)** Explain how the antithetic variate technique works in reducing the variance of a Monte Carlo estimate. When is this technique effective to reduce the variance? Does it provide an unbiased estimate?:
3. **(8 points)** Consider a foreign market and a domestic market. A log-normal risky stock and a riskless bond trade in each market (i.e. we have a domestic bond, a domestic stock, a foreign bond and a foreign stock). Assume the exchange rate is lognormal as well. Derive the dynamics of the exchange rate and of the foreign risky security with respect to the domestic risk neutral measure. Derive also the covered interest rate parity that relates the forward exchange rate to the spot exchange rate.
4. **(7 points)** State Ito formula in the jump diffusion setting (specify all the assumptions on the involved stochastic processes).
5. **(5 points)** Let S be the risky security in the jump diffusion setting. Compute the stochastic differential of

$$Y(t) = (S(t))^{\frac{1}{3}}$$

with respect to *the historical* and the *risk-neutral probability measure* using Ito formula stated in the previous point

Solution of the Exercises

1. Given the function

$$f(t, S) = (S)^{\frac{1}{3}}$$

the derivatives are

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial S} = \frac{1}{3} \cdot (S)^{-\frac{2}{3}}, \quad \frac{\partial^2 f}{\partial S^2} = -\frac{2}{9} \cdot (S)^{-\frac{5}{3}}$$

and therefore

$$\begin{aligned} dY(t) &= 0dt + \frac{1}{3} \cdot (S(t^-))^{-\frac{2}{3}} S(t^-) (\mu dt + \sigma dW(t)) + \\ &\quad + \frac{1}{2} \left(-\frac{2}{9} \cdot (S(t^-))^{-\frac{5}{3}} \right) \sigma^2 S^2(t^-) dt + \\ &\quad + \left((S(t))^{\frac{1}{3}} - (S(t^-))^{\frac{1}{3}} \right) \cdot \Delta N(t) \end{aligned}$$

Hence

$$\begin{aligned} dY(t) &= Y(t^-) \left(\frac{1}{3} (\mu dt + \sigma dW(t)) - \frac{1}{9} \sigma^2 dt \right) + \\ &\quad + (Y(t) - Y(t^-)) \cdot \Delta N(t) \end{aligned}$$

Since

$$S(t) = S(t^-) Y_j \text{ if } \Delta N(t) = 1 \text{ with } j = N(t)$$

we have that

$$Y(t) = S(t)^{\frac{1}{3}} = S(t^-)^{\frac{1}{3}} Y_j^{\frac{1}{3}} = Y(t^-) Y_j^{\frac{1}{3}} \text{ if } \Delta N(t) = 1 \text{ with } j = N(t)$$

and thus

$$\frac{dY(t)}{Y(t^-)} = \left(\frac{1}{3} \mu - \frac{1}{9} \sigma^2 \right) dt + \frac{1}{3} \sigma dW(t) dt + \left(Y_j^{\frac{1}{3}} - 1 \right) \cdot \Delta N(t)$$

is the stochastic differential equation of Y w.r.t. the historical probability measure. Moving under the risk neutral one, we observe that

$$\begin{aligned} \frac{dS(t)}{S(t^-)} &= r dt + \sigma dW_t^Q + (\Delta J(t) - (\mu_Y - 1) \lambda^Q dt) = \\ &= (r - (\mu_Y - 1) \phi \lambda^P) dt + \sigma dW_t^Q + \Delta J(t) \end{aligned}$$

implies

$$\frac{dY(t)}{Y(t^-)} = \left(\frac{1}{3} \underbrace{(r - (\mu_Y - 1)\phi\lambda^P)}_{\text{instead of the historical drift } \mu} - \frac{1}{9}\sigma^2 \right) dt + \frac{1}{3}\sigma dW^Q(t) dt + \left(Y_j^{\frac{1}{3}} - 1\right) \cdot \Delta N(t)$$

where W_t^Q is the standard Q -Brownian motion. The Poisson process N preserves the same distribution w.r.t. Q with risk neutral intensity $\phi\lambda^P$. The sequence Y_j is i.i.d. with the same distribution both with respect to P and Q . All processes W^Q , N and the sequence of Y_j are Q -independent.

Applied Numerical Finance

Code 20247

Bocconi University

Final Exam

September 2015

Instructions. The exam lasts 60 minutes and counts for 31 points.

Good luck!

Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate. Suppose that $S_1(0) = S_2(0)$.

1. **(7 points)** The *terminal payoff* of a European derivative with maturity T is

$$X(T) = \begin{cases} 1 & \text{if } S_2(T) < S_1(T) \\ 0 & \text{if } S_2(T) \geq S_1(T) \end{cases}$$

Determine its no-arbitrage price at $t = 0$.

2. **(6 points)** Compute the replicating strategy of the derivative of the previous point at $t = 0$. Is the replicating strategy initially long/short on B , S_1 and S_2 ?
3. **(6 points)** Define the forward-difference estimator and the central-difference estimator for the sensitivities of a given derivative. Are these estimators unbiased? Which is better? Justify your answer.
4. **(4 points)** State the backward recursion to compute the price of an American put option in discrete time.
5. **(8 points)** Explain (with details) how the control variate technique works in reducing the variance of a Monte Carlo estimate. When is this technique effective to reduce the variance? Does it provide an *unbiased* estimate?

1. The no-arbitrage price at $t = 0$ of the European derivative whose payoff at maturity T is

$$X(T) = \begin{cases} 1 & \text{if } S_2(T) < S_1(T) \\ 0 & \text{if } S_2(T) \geq S_1(T) \end{cases}$$

is given by

$$V_X(0) = \mathbb{E}^{\mathbb{Q}} [\mathbb{I}_{\{S_2(T) < S_1(T)\}} e^{-rT}] = \mathbb{Q}[S_2(T) < S_1(T)] e^{-rT}.$$

We first compute

$$\begin{aligned} & \mathbb{Q}[S_2(T) < S_1(T)] = \\ &= \mathbb{Q} \left[S_2(0) \exp \left(\left(r - \frac{1}{2} \|\sigma_2\|^2 \right) T + [\sigma_{21}, \sigma_{22}] \cdot W^{\mathbb{Q}}(T) \right) < S_1(0) \exp \left(\left(r - \frac{1}{2} \|\sigma_1\|^2 \right) T + [\sigma_{11}, \sigma_{12}] W_1^{\mathbb{Q}}(T) \right) \right] \\ &= \mathbb{Q} \left[([\sigma_{21}, \sigma_{22}] - [\sigma_{11}, \sigma_{12}]) \cdot W^{\mathbb{Q}}(T) < \ln \frac{S_1(0)}{S_2(0)} - \frac{1}{2} (\|\sigma_1\|^2 - \|\sigma_2\|^2) T \right] \end{aligned}$$

where $W^{\mathbb{Q}} = [W_1^{\mathbb{Q}}, W_2^{\mathbb{Q}}]^T$ is the standard 2-dimensional risk neutral Brownian motion. Since

$$([\sigma_{21}, \sigma_{22}] - [\sigma_{11}, \sigma_{12}]) \cdot W^{\mathbb{Q}}(T) \stackrel{\mathbb{Q}}{\sim} \left\| \begin{bmatrix} \rho v_2 - v_1 \\ \sqrt{1 - \rho^2} v_2 \end{bmatrix} \right\| N(0, T)$$

and

$$\begin{aligned} \left\| \begin{bmatrix} \rho v_2 - v_1 \\ \sqrt{1 - \rho^2} v_2 \end{bmatrix} \right\| &= \sqrt{(\rho v_2 - v_1)^2 + (\sqrt{1 - \rho^2} v_2)^2} = \\ &= \sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \end{aligned}$$

we get

$$([\sigma_{21}, \sigma_{22}] - [\sigma_{11}, \sigma_{12}]) \cdot W^{\mathbb{Q}}(T) \stackrel{\mathbb{Q}}{\sim} \sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T} Z^{\mathbb{Q}}$$

with $Z^{\mathbb{Q}} \stackrel{\mathbb{Q}}{\sim} N(0, 1)$. Therefore,

$$\begin{aligned} \mathbb{Q}[S_2(T) < S_1(T)] &= \mathbb{Q} \left[\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T} Z^{\mathbb{Q}} < \ln \frac{S_1(0)}{S_2(0)} - \frac{1}{2} (\|\sigma_1\|^2 - \|\sigma_2\|^2) T \right] \\ &= \mathbb{Q} \left[Z^{\mathbb{Q}} < \left(\ln \frac{S_1(0)}{S_2(0)} - \frac{1}{2} (v_1^2 - v_2^2) T \right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right] \\ &= N \left(\left(\ln \frac{S_1(0)}{S_2(0)} - \frac{1}{2} (v_1^2 - v_2^2) T \right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right) \end{aligned}$$

Therefore the initial no-arbitrage price of the derivative is

$$\begin{aligned} V_X(0) &= e^{-rT} N \left(\left(\ln \frac{S_1(0)}{S_2(0)} - \frac{1}{2} (v_1^2 - v_2^2) T \right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right) \\ &= e^{-rT} N \left(\left(0 - \frac{1}{2} (v_1^2 - v_2^2) T \right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right), \end{aligned}$$

exploiting the assumption $\frac{S_1(0)}{S_2(0)} = 1$.

2. The components of the replicating self-financing strategy of the derivative at $t = 0$ are

$$\begin{aligned}\vartheta_0(0) &= V_X(0) - \vartheta_1(0) S_1(0) - \vartheta_2(0) S_2(0) \\ \vartheta_1(0) &= \frac{\partial}{\partial S_1(0)} V_X(0) \\ \vartheta_2(0) &= \frac{\partial}{\partial S_2(0)} V_X(0)\end{aligned}$$

In order to compute

$$\vartheta_1(0) = \frac{\partial}{\partial S_1(0)} V_X(0)$$

we recall from our previous computations that

$$V_X(0) = N(\xi) e^{-rT}$$

where

$$\xi = \left(\ln \frac{S_1(0)}{S_2(0)} - \frac{1}{2} (v_1^2 - v_2^2) T \right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}$$

and therefore

$$\begin{aligned}\frac{\partial}{\partial S_1(0)} V_X(0) &= e^{-rT} \frac{\partial}{\partial S_1(0)} N(\xi) \\ &= e^{-rT} f_{N(0,1)}(\xi) \frac{\partial \xi}{\partial S_1(0)}\end{aligned}$$

where $f_{N(0,1)}$ denotes the probability density of a standard normal random variable. Since

$$\frac{\partial \xi}{\partial S_1(0)} = \frac{1}{S_1(0)} \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}$$

we obtain

$$\vartheta_1(0) = \frac{\partial}{\partial S_1(0)} V_X(0) = e^{-rT} f_{N(0,1)}(\xi) \left(\frac{1}{S_1(0)} \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right) > 0$$

Similarly,

$$\begin{aligned}\vartheta_2(0) &= \frac{\partial}{\partial S_2(0)} V_X(0) = e^{-rT} \frac{\partial}{\partial S_2(0)} N(\xi) \\ &= e^{-rT} f_{N(0,1)}(\xi) \frac{\partial \xi}{\partial S_2(0)}\end{aligned}$$

since

$$\frac{\partial \xi}{\partial S_2(0)} = -\frac{1}{S_2(0)} \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}$$

we obtain

$$\vartheta_2(0) = -e^{-rT} f_{N(0,1)}(\xi) \frac{1}{S_2(0)} \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} < 0$$

And finally

$$\begin{aligned}\vartheta_0(0) &= V_X(0) - \vartheta_1(0) S_1(0) - \vartheta_2(0) S_2(0) \\ &= N(\xi) e^{-rT} - e^{-rT} f_{N(0,1)}(\xi) \left(\frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right) - e^{-rT} f_{N(0,1)}(\xi) \left(-\frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right) \\ &= N(\xi) e^{-rT} > 0\end{aligned}$$

Hence the replicating strategy of the derivative is initially long on B and S_1 and short on S_2 .